

LIQUIDITY – PROFITABILITY TRADEOFF: AN EMPIRICAL INVESTIGATION IN AN EMERGING MARKET

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This study empirically examines the relation between profitability and liquidity, as measured by current ratio and cash gap (cash conversion cycle) on a sample of joint stock companies in Saudi Arabia. Using correlation and regression analysis the study found significant negative relation between the firm's profitability and its liquidity level, as measured by current ratio. This relationship is more evident in firms with high current ratios and longer cash conversion cycles. At the industry level, however, the study found that the cash conversion cycle or the cash gap is of more importance as a measure of liquidity than current ratio that affects profitability. The size variable is also found to have significant effect on profitability at the industry level. Finally, the results are stable over the period under study.

INTRODUCTION

Efficient liquidity management involves planning and controlling current assets and current liabilities in such a manner that eliminates the risk of the inability to meet due short-term obligations, on one hand, and avoids excessive investment in these assets, on the other. This is due in part to the reduction of the probability of running out of cash in the presence of liquid assets.

The working capital approach to liquidity management has long been the prominent technique used to plan and control liquidity. The working capital includes all the items shown on a company's balance sheet as short-term or current assets, while net working capital excludes current liabilities. This measure is considered a useful tool in accessing the availability of funds to meet current operations of companies. However, instead of using working capital as a measure of liquidity, many analysts advocate the use of current and quick ratios, which have the advantage of making temporal or cross sectional comparison possible.

However, the ultimate measure of the efficiency of liquidity planning and control is the effect it has on profits and shareholders' value. Thus, this study attempts to examine the relation between liquidity and profitability using a sample of Saudi joint stock companies. Second, the study aims at directing the attention to the importance of active management of liquidity. This aspect is more important given the number of non-profitable Saudi companies, and the dire need to improve profitability.

To carry out these objectives the remainder of this paper is organized as follows: the next section reviews the literature for relevant theoretical and empirical work on liquidity and cash management and its effect on profitability. Section three describes the sample and the methodology followed in this study. Section four portrays and discusses the statistical results,

while section five explores the implications of the study. The final section, section six, concludes the paper.

LITERATURE REVIEW

Working capital represents a safety cushion for providers of short-term funds of the company, and as such they view positively the availability of excessive levels of working capital and cash. However, from an operating point of view, working capital has increasingly been looked at as a restraint on financial performance, since these assets do not contribute to return on equity (Sanger, 2001). Furthermore, liquidity management is important in good times and it takes further importance in troubled times. The efficient management of the broader measure of liquidity, working capital, and its narrower measure, cash, are both important for a company's profitability and well being. In the words of Fraser (1998) "there may be no more financial discipline that is more important, more misunderstood, and more often overlooked than cash management." However, as argued vividly by Nicholas (1991,) companies usually do not think about improving liquidity management before reaching crisis conditions or becoming on the verge of bankruptcy.

Survey of working capital and cash management literature, however, shows that instead of linking liquidity and cash management to a known efficiency or profitability measures, the majority of research, especially the earlier efforts, attempts to develop models for optimal liquidity and cash balances, given the organization's cash flows. The earlier cash management research focused on using quantitative models that weight the benefits and costs of holding cash (liquidity). Under this category falls Baumol's (1952) inventory management model and Miller and Orr's (1966) model which recognizes the dynamics of cash flows. The benefit of these earlier models is that they help financial managers understand the problem of cash management, but they do require assumptions that may not hold in practice.

Similarly, Johnson and Aggarwal (1988) support a treasury approach to cash management, which concentrates on flows which entail that cash collection and payment cycles must be broken into their constituent parts. Management then should review the time needed for each link in the collection and payment cycles. Some policy outlines, similar to these, were proposed by Schneider (1988) by arguing that cash management should include analytical review of the procedures followed in managing working capital. These include granting of credit, managing balances, and collecting payments when due.

The subsequent and more practical approaches to liquidity management focused on working capital requirements and levels of desired liquidity as measured by current ratio and its variants. The finance textbooks and literature covered the techniques and approaches aimed at managing working capital and its individual components. Again most of these approaches attempt to develop an optimal level of working capital components under certain assumptions, albeit less restrictive than their earlier counterparts used to facilitate development of cash management techniques.

Various other techniques have been suggested to improve liquidity and cash positions and to increase the efficiency of their management and in turn profitability. These include credit insurance (Brealey and Myers, 1996; Unsworth, 2000; and Raspanti, 2000), factoring of receivables (Brealey and Myers, 1996; Summers and Wilson, 2000).

However, as measures of liquidity, both the working capital and liquidity ratios have come under criticism for various reasons. Hawawini et al. (1986), for example, argue that the

concept of working capital requirement is a better measure of a firm's investment in its operating cycle than the traditional concept of net working capital. Similarly, Finnerty (1993) points out that the traditional liquidity ratios, such as the current ratio or quick ratio, include both liquid financial assets and operating assets in their formula. Thus, from an ongoing concern point of view, the inclusion of operating assets which are tied up in operations is not useful. Kamath (1989), however, argues that both current and quick ratios are deficient due to their static nature and the inadequacy of using them as measures of future cash flows and liquidity.

These shortcomings of working capital and liquidity ratios have led researchers and analysts to advocate other measures of liquidity that are more indicative of cash availability. The net cash conversion cycle, or the cash gap has been suggested by many as a possible supplement or replacement to the working capital and current ratios as measures of available liquidity (see Gitman (1974), Richard and Laughlin (1980), Boer (1999), and Gentry et al., (1990).

It is generally argued that this approach is more practical due to the dynamic nature of cash cycles and the many complications and tradeoffs involved. Some authors, such as Kamath (1989), suggest that cash gaps can be used to replace or supplement liquidity ratios (current ratio and quick ratio) in measuring and predicting the nature and pattern of future cash flows. The static current ratios alone fall short of adequately predicting future cash flows. Other studies, such as Kolay (1991), differentiate between short-term and long-term strategies that improve financial position and cash management policies.

The cash gap, known also as cash flow cycle or cash conversion cycle, measures the length of time between actual cash expenditures on productive resources and actual cash receipts from the sale of products or services. Thus, this definition for cash conversion cycle indicates that a shorter cash cycle or gap is desirable since the larger the cash cycle or gap the greater the need for external financing and the greater the financing costs to be borne in form of explicit interest costs or implicit costs of other financing sources, such as equity. The interest cost is more expensive in Saudi Arabia, than in other countries, because of the absence of tax savings (national companies incorporated in Saudi Arabia are not required to pay taxes, they instead pay zakat (level or fixed percentage tax required by Islamic sharia).

To emphasize the importance of managing liquidity, Loeser (1988) tapped the extreme in order to reduce the cash cycle. Loeser recommended assessing interest charge at the prime rate to outstanding accounts receivable and unbilled revenue in order to encourage responsible employees and departments within companies to put every effort necessary to collect receivables, and thus reduce cash gaps. This same approach was expressed recently by Fraser (1998) who argues that liquidity and cash gap management starts with a simple task for financial managers by making certain that their billings, collections, and payables systems are operating efficiently.

The direct effect of liquidity is not only on the cash position and the troubles it may cause to financial managers, but it rather effects the company's profits in a more direct way. This direct effect stems from the need of the company to borrow to finance the working capital requirements and cash gaps. For example, if a company has a cash gap of 100 days, this means that the company has to borrow an amount equivalent to 100 times the daily cost of sales. The borrowing cost reduces both pretax and after-tax profits by equal amounts. In Saudi Arabia the feature of borrowing cost as a cheap source of financing loses its tax advantage since there is no tax on Saudi companies' profits. Likewise, reducing cash gaps by any number of days will add equally to the pretax and after-tax profits.

Shin and Soenen (1998) investigated the relation between the firm's net trade cycle and its profitability, using a large sample of American firms during 1975-1994. The study found a

strong negative relation between the length of the firm's net trade cycle and various measures of profitability, including market measures, such as stock returns, and operating profits. Similarly, this study attempts to examine the relationship between operating profitability and liquidity measures. Unlike previous studies, an attempt is made here to study the effects of various levels of liquidity, in its broader or narrow sense, on a company's profitability.

DATA AND METHODOLOGY

Since the aim of this study is to examine the relation between profitability and liquidity, the study makes a set of testable hypotheses. First, this study assumes that there may be a relationship between profitability of the company and its liquidity profile, since the later effects the former in a direct way, as a result of the external financing costs or savings thereof. Due to these elements of costs and cost savings this relationship is most likely be negative. Thus, the first hypothesis of this study can be stated as follows:

Hypothesis 1

There is a possible negative relation between liquidity of a company and its profitability. Companies with relatively high levels of liquidity are expected to post low levels of profitability and vice versa.

Secondly, profitability, on the other hand, may be a function of the size of companies (measured in terms of sales or total assets). The company size may affect liquidity, cash gaps and, hence, profitability in different ways. On the one hand, large companies may be able to buy inventory in large quantities in order to get quantity discounts. Further, because of their size, large companies may qualify for quantity discounts from suppliers with relatively small inventory levels. On the other hand, large companies may be able to get favorable credit terms from their suppliers in terms of longer credit periods. Moreover, large companies may have more success in their receivables collection efforts relative to small companies. All these factors may push liquidity levels and cash gaps of large companies to levels lower than that of small companies. On the contrary, small companies are usually not able to obtain as much inventory to qualify for quantity discounts as their large counterparts do. Additionally, small companies make efforts to pay within discount periods in order to benefit from cash discounts and to avoid severing their relations with their suppliers. These factors may force small companies to have higher liquidity levels and larger cash gaps. Accordingly, this study states the following hypothesis:

Hypothesis 2:

A positive relation may exist between the company size and its profitability. This may be due to the ability of large companies to reduce liquidity levels and cash gaps.

Third, liquidity and cash gaps may differ among industries and among countries and may depend on the prevailing economic conditions. Sometimes traditions and the nature of business set the typical working capital requirements and the cash gap in a given industry. Some industries have inherently high levels of working capital requirements and large cash gaps than others, while some may require low levels of working capital and shorter or even negative cash gaps, which indicate their ability to obtain cost-free capital from their customers. Hawawini et al. (1986) examined a sample of 1181 American firms from thirty-six industries over a period of nineteen years and found significant and persistent industry effects on a firm's investment in working capital. The ability to operate with low levels of working capital and obtaining cost-free

capital may have direct positive bearing on profitability. Thus, this study states the following hypothesis:

Hypothesis III

Need for working capital and liquidity is influenced by the industry in which the company operates. Capital intensive industries require low levels of working capital and tend to have smaller cash gaps than their labor-intensive counterparts. Accordingly, liquidity requirement is expected to have no significant negative impact on profitability of capital - intensive industries, while such effect is expected in labor- intensive ones.

To test these hypotheses this study uses the following methodology:

1. The study first estimates the cash gap for each company and for each year of the sample period as follows:

$$\text{Cash Gap} = \text{Days in Inventory (DII)} + \text{Days in Accounts Receivable (DIR)} - \text{Days in Accounts Payable (DIP)}$$

The components of the cash gap are calculated as follows:

- Inventory turnover = cost of goods sold/ average inventory
- Number of days in inventory= 365/ inventory turnover
- Number of days in Receivables= Receivables/ average daily sales
- Number of days in Payables = Payables / average daily purchases

2. Correlation analysis to identify the association between profitability and liquidity indicators and other related variables
3. Regression analysis to estimate the causal relationship between profitability variable, liquidity and other chosen variables

The data for this study comes from a sample of Saudi joint stock companies. Overall, 29 joint stock companies that are publicly traded and provide annual audited financial reports are selected. This sample encompasses three basic Saudi economic sectors. Table 1 shows the sample by sector over the period 1996-2000. However, due to unavailability of data in some of the years for some companies and sectors, the distribution of the sample is not homogenous over the sample period.

TABLE 1

Sample Distribution and Description

Economic Sector	1996	1997	1998	1999	2000
Agriculture	5	7	7	6	5
Industrial	11	10	12	8	8
Services	5	7	8	8	-
Total	21	24	27	22	13

The sample does not include electricity and banking sector companies. The former is regulated and has undergone major structural changes during the sample period, while the banking sector activity does not fit the issues at hand. It should be mentioned that some problems are encountered in collecting the data for this study. First, most cement companies in the Kingdom do not disclose sales revenue. Second, most companies do not report the purchases figure or detailed cost of goods sold figures. Thus, the sample is restricted to those companies

for which purchases figure can be calculated and the sales figures are available. Both components are necessary to calculate cash gaps. Nevertheless, the total sample represents about 50 percent of the total number of Saudi publicly held companies (excluding Banking and electricity companies). Thus, the final sample includes the most important joint stock companies in Saudi Arabia (See Appendix A to this study for a list of the companies included in the sample).

RESULTS AND ANALYSIS

The following notations are used throughout this study:

S= net sales

TA= Total assets

CG= Cash gap in days

CR= Current ratio

LOGS= Logarithm of net sales

LOGTA= Logarithm of total assets

CGS= Cash gap in days/100

NOI= Net operating income + depreciation / net sales

Table 2 shows the measures of central tendency and dispersion for the basic variables used in this study; NOI, CR, CG, and S. The table shows wide variation for these variables, especially Sales (S) and Cash gap (CG). Thus these two variables are transformed in the analysis that follows, by taking the logarithm of sales (to satisfy normality) and divide CG by 100, to get a new scaled variable CGS.

TABLE 2

Descriptive Statistics (n =107)

	S(mills)	CG (days)	CR	NOI
Mean	2882.08	180	2.835	.1936
Std Dev.	20336	184	5.22	.256
Minimum	7.9	-592	.4	-.55
Maximum	208333.2	948	34.1	1.29
Q1	107.6	88	1.1	.078
Q3	550.08	247	2.1	.326

The study first examines the relationship that exists between the variables of profitability and liquidity. Table 3 shows the Pearson correlation coefficients between NOI, CGS, LOGTA, LOGS, CR, CG, and TA for the whole sample of 107 company-year observations.

TABLE 3
Pearson Correlation Coefficients Matrix

	TA	CG	CR	LOGS	LOGTA	CGS	NOI
S	.571* (.00)	-.071 (.465)	-.026 (.791)	.507* (.000)	.430* (.000)	-.071 (.465)	-.041 (.673)
TA	1.00	-.101 (.30)	-.057 (.561)	.655* (.00)	.723* (.00)	-.101 (.30)	.08 (.411)
CG		1.00 (.00)	.001 (.99)	-.036 (.71)	-.085 (.385)	1.00* (.00)	-.124 (.20)
CR			1.00 (.00)	-.388* (.00)	-.217** (.025)	.001 (.99)	-.379* (.00)
LOGS				1.00 (.00)	-.868* (.00)	-.036 (.71)	.205** (.034)
LOGTA					1.00	-.085 (.385)	.262* (.006)
CGS						1.00 (.00)	-.124 (.202)

- * indicates significance at all levels
- ** indicates significance at .05 level

If efficient liquidity management increases profitability, one should expect a negative relationship between the measures of liquidity and profitability variable. In Table 3 the correlation coefficients indicate a significant negative relationship between current ratio (CR) and Net Operating Income (NOI). However, the relationship between NOI and Cash Gap (CG/CGS) is negative, but is not statistically significant. These results indicate that these two variables (CR and CG) measure liquidity differently.

One should not overlook the positive significant association that exists between NOI and the two measures of size, LOGS and LOGTA, which in turn indicates a positive relation between size and profitability. This has to be viewed in light of the highly significant negative correlation that exists between these two variables and the CR. Similarly, a negative relation exists between LOGS and LOGTA and CG, but is not significant. Finally, one should observe the strong highly positive relation that exists between sales and total assets which show that they are substitute measures of size.

To investigate the association between profitability and liquidity further, the study estimates the following regression equation for the whole sample and for four sub- samples. The sub-samples are by products of dividing the sample into four sub- samples, based on CR and CG. The first two sub- samples are created by dividing the sample into two sub- samples, one with $CR \leq 2$ and the other with $CR > 2$. The reason for this split is to examine whether the effect on profitability is a function of the level of liquidity. The other two samples are created by dividing the sample into two sub- samples, one with $CG \leq 150$ days and the other with $CG > 150$ days. The purpose in this case is to examine whether the relation between profitability, liquidity level

and cash gap is a function of the efficiency of managing cash cycle. For the sample and each of the sub- samples the following regression equation is estimated:

NOI= B₀+B₁ CR+B₂CGS+B₃LOGS+e

Where:

B₀, B₁, B₂ B₃, the coefficients of the regression equation; and the other variables are as defined before.

TABLE 4

NOI= B₀+B₁ CR+B₂CGS+B₃LOGS+e
Regression (1): The general regressions

	Intercept	CR	CGS	LOGS	R ²	F	N
<u>Pooled</u> B t Tolerance VIF	2.18 ** (2.2)	-.017* (3.63) .849 1.178	-.017 (1.35) .998 1.002	.010 (.639) .848 1.179	.163	6.66*	107
<u>Sub- sample CR< 2</u> B t Tolerance VIF	.414 (.139)	-.087 (1.23) .912 1.097	-.023 (1.52) .928 1.077	-.003 (.158) .97 1.031	.071	1.73 □□□	72
<u>Sub-sample CR> 2</u> B t Tolerance VIF	-.192 (1.36)	-.009* (2.21) .653 1.531	-.034 (1.57) .999 1.00	-.059* (2.44) .654 1.53	.486	9.76*	35
<u>Sub- sample CG<150</u> B t Tolerance VIF	1.48 * (3.06)	-.0109 (1.59) .927 1.079	-.11* (4.69) .965 1.036	-.0219 (1.32) .944 1.059	.422	9.96*	45
<u>Sub- sample CG>150</u> B t Tolerance VIF	.198 (1.08)	-.018* (2.684) .644 1.553	-.005 (.225) .961 1.04	-.0076 (.241) .643 1.556	.182	4.28*	62

VIF= Variance Inflation Factor.
(*) indicates significance at all levels

Table 4 shows the results of estimating the regression equation for the pooled sample and for each of the four sub-samples. The results of the pooled regression and the sub-samples, in general, show that the model fits the data very well shown by the high R^2 and F statistics. However, the pooled regression shows that the current ratio with a negative and very significant slope coefficient confirming the results of the earlier correlation analysis. Similarly, the coefficient for the CGS is negative, but insignificant. The size variable LOGS is positive, but also not significant.

The sub-sample regressions, however, show different results. For the sub-sample with $CR \leq 2$ the coefficients of CR and CGS are both negative, but are insignificant and so is LOGS. This indicates that when liquidity levels are low, denoted by low current ratios, the effect on profitability is not significant. However, when $CR > 2$, there is highly negative effect on profitability. This effect is multiplied when the company size is large, evidenced by the significant negative coefficients of LOGS.

Similarly, when the sample is divided on the basis of cash gap, the results for the sub-samples where $CG \leq 150$ show that CR has a negative, but insignificant coefficients indicating insignificant effect on profitability. This means that when the cash conversion cycle is short the current ratio loses its importance and does not impact profitability. However, since the cash conversion cycle (cash gap) replaces the current ratio as a liquidity parameter, the cash gap coefficient is negative and highly significant pointing to the negative impact of relatively large cash gaps on profitability. When the cash gap is greater than 150 days the CR coefficient is negative and highly significant. This indicates that when the cash conversion cycle is long, then the liquidity level (as measured by current ratio) becomes a significant parameter and impacts profitability in a significantly negative way.

However, it is highly possible that the independent variables, CR, CGS, and LOGS are highly related to each other, and inclusion of all of them will thus be unjustified. To test for this possibility the study used two methods recommended by econometricians, which are Tolerance and Variance Inflation Factor (VIF) tests (see Gujarati, 1995). The tolerance is a statistical measure used to determine how much the independent variables are linearly related to each other. A variable with very low tolerance contributes little information to the model. It is measured as $(1-r^2)$ for the variable, or simply as a reciprocal of the Variable Inflation factor, $1/VIF$. When the variable is highly collinear the tolerance level is zero or close to zero. If, on the other hand, it is not related to other variables the tolerance level is 1. The VIF, on the other hand, is a measure that is used to determine collinearity. Large VIF value is an indication of multicollinearity. As a rule of thumb, if VIF of a variable exceeds 10 that variable is said to be highly collinear. It is evident from Table 4 that the independent variables are independent and there is no collinearity among them. Thus, the regression results together with the correlation results support the first hypothesis of this study on the overall negative relation between profitability and liquidity. It is evident, however, that this effect depends on the level of liquidity and the size of cash gap.

On the other hand, the sector regressions show different patterns than the pooled regression. For the Agricultural and manufacturing sectors both CGS and LOGS coefficients are positive and highly significant, while the CR coefficient is not significant. This may explain the relative importance of the cash gap over current ratio as a measure of liquidity, in the capital-intensive industries such as Agricultural and Industrial sectors. Similarly within each sector, large size companies are usually more able to tolerate larger cash gaps than smaller ones. Thus, the results confirm hypothesis II and III on the influence of both industry and size on profitability within economic sectors. In the Service sector, however, cash gap loses its importance since it is

known that the cash gaps for companies within the service sector are relatively small. Hence it is the level of current assets and current liabilities that impact profitability. Also, the company size has no effect on profitability, since it is the efficiency of using assets rather than size of assets that affects profitability in the service sector.

TABLE 5

$$\text{NOI} = B_0 + B_1 \text{CR} + B_2 \text{CGS} + B_3 \text{LOGS} + e$$

Regression (2): Sector Regressions

	Intercept	CR	CGS	LOGS	R ²	F	N
<u>Agriculture</u>							30
B	-.773 *	-.0043	.178 *	.136 *	.679	18.36 *	
t	(4.08)	(1.024)	(4.63)	(4.24)			
Tolerance		.555	.932	.527			
VIF		1.802	1.073	1.899			
<u>Industrial</u>							54
B	-.175	.026	.0226**	.0234*	.225	4.06 **	
t	(1.75)	(1.35)	(2.63)	(2.95)			
Tolerance		.961	.842	.87			
VIF		1.04	1.187	1.15			
<u>Services</u>							23
B	1.48 *	-.425 *	.0036	-.114	.503	6.41 *	
t	(3.06)	(3.92)	(.117)	(1.46)			
Tolerance		.575	.829	.669			
VIF		1.738	1.207	1.507			

VIF= Variance Inflation Factor, (*) indicates significance at all levels

(**) indicates significance at .05 level

There is an important question concerning the stability of the regression results over time. To examine this issue a regression is conducted for each individual year in the sample period. However, due to the sheer number of observations in 2000, the regression for 2000 is done by combining 1999 and 2000 observations together. The results shown in Table 6 for the individual years further support the assumed relations of this study and their stability over time. First, the high R² shows that a large variation of the dependent variable is explained by the model. The size coefficient for all years is not significant indicating that size matters when considered within sectors or industries. The CR coefficients for all years are all negative as expected and significant for most of the time period, while CGS coefficients are negative for most of the years but insignificant.

TABLE 6

$NOI = B_0 + B_1 CR + B_2 CGS + B_3 LOGS + e$
Regression (3): Temporal Regression

	Intercept	CR	CGS	LOGS	R ²	N
<u>1996:</u> B t Tolerance VIF	.494 * (2.97)	-.106* (2.96) .929 1.077	-.029** (2.02) .996 1.004	-.0141 (.618) .927 1.079	.42	21
<u>1997:</u> B t Tolerance VIF	.056 (.262)	-.010 (1.24) .887 1.127	-.025 (1.012) .999 1.001	.0148 (1.15) .886 1.129	.208	24
<u>1998:</u> B t Tolerance VIF	.15 (.66)	-.022** (1.97) .781 1.281	-.0024 (.553) .982 1.018	.021 (.553) .791 1.264	.224	27
<u>1999:</u> B t Tolerance VIF	.477 (1.32)	-.019*** (1.82) .711 1.4	.0031 (.056) .882 1.134	.0323 (.643) .666 1.502	.167	22
<u>1999/2000</u> B t Tolerance VIF	.216 (.85)	-.0184* (2.02) .685 1.456	.031 (.76) .911 1.098	-.0044 (.125) .661 1.5137	.176	35

(*) indicates significance at all levels
(**) indicates significance at .05 level
(***) indicates significance at .10 level

IMPLICATIONS OF THE STUDY

The results of this study have important implications for liquidity management in various Saudi companies. First, it is clear that there is a negative relation between profitability and liquidity indicators such as current ratio and cash gap in the Saudi sample examined. The study also revealed that there is great variation among industries with respect to the significant measure of liquidity. Although certain liquidity levels are desirable and sometimes unavoidable, the study points to the lost profits and the unnecessary costs that are borne by companies as a result of holding excessive liquidity. These losses or costs could be reduced or eliminated by adopting active liquidity management strategies.

Furthermore, the study highlights and reviews briefly the importance of using the various measures and techniques that contribute towards an efficient management of liquidity and, hence, improve profitability. These techniques include the JIT system, credit insurance, and factoring of receivables, to mention a few.

SUMMARY AND CONCLUSIONS

This study has attempted to empirically examine the relation between profitability and liquidity in a sample of 29 joint stock companies that represent the major economic sectors in the kingdom (excluding the Electricity and Banking industries) over the period 1996-2000. The study reveals that there exists a significant and negative relation between profitability and liquidity measures such as current ratio and cash gap. The study found that CR is the most important liquidity measure that affects profitability. This effect, however, varies with the level of liquidity as measured by CR. However, within sectors the cash gap, as a liquidity measure, is found to be more important than CR in affecting profitability. Size is also found to bear some influence over profitability within economic sectors, but not in the overall sample. These two influences are more profound in capital- intensive sectors, such as manufacturing and agriculture. Nevertheless, cash gap is found to lose its importance within the labor- intensive sectors, such as services.

APPENDIX A

The Companies included in the Sample (by Sector)

Agricultural:

Saudi Fisheries Company
National Agricultural Development Company
Tabuk Agricultural Development Company (TADCO)
Aljouf Agricultural Development Company
Jizan Agricultural Development Company (GAZADCO)
Hail Agricultural Development Company (HADCO)
Qassim Agricultural Development Company (GACO)
Ash-Sharqia Agricultural Development Company (SHADCO)

Industrial

National Gypsum Company
Saudi Ceramic Company
Arabian Pipes Company
Food Products Company
Saudi Company for Industrial Development (SIDC)
Saudi Pharmaceutical Industries and Medical Application Company (SPIMACO)
Saudi Basic Industries Corporation (SABIC)
The Saudi Arabia Aminantit Company
National Industrialization Company
Saudi Cable Company
Saudi Arabian Fertilizers Company (SAFCO)
National Gas and Industrialization Company (GASCO)

The Savola Group

Services

Saudi Arabian Public Transport Company (SAPTCO)
Saudi Automatic Services Company (SASCO)
Saudi Hotels and Resort Areas Company (SHARACO)
Al-mawashi and Al-mukairish United
Makkah Construction and Development Company
Saudi Real Estate Company (SRECO)
National Agricultural Marketing Company (THIMAR)
Saudi Land Transport Company (Mubbarrad)

REFERENCES

- Baumol, W.J. (November, 1952). The transaction demand for cash: An inventory theoretic approach. *Quarterly Journal of Economics*, 66, 545-556.
- Brealey, Richard A. and Stewart C. Myers (1996). *Principles of corporate finance*. McGraw-Hill.
- Brigham, Eugene F., and Louis C. Gapenski (1987). *Intermediate financial management*. Dryden Press.
- Boer, Germain. (1999, November). Managing the cash gap. *Journal of Accountancy*, 27-32.
- Bort, Richard. (1987, Sept/Oct). Managing the treasury function: An update. *Journal of Cash Management*, 7(5), 14-15.
- Finnerty, J. E. (1993). *Planning cash flow*. American Management Association.
- Fraser, Jill A. (1998, October). The art of cash management. *Inc* (pp. 124-125).
- Gentry, James A., R. Vaidyanathan, and Hei Wai Lee. (1990, Spring). A weighted cash conversion cycle. *Financial Management*, 19, 90-94.
- Gitman, Laurence J. (1974). Estimated corporate liquidity requirements: A simplified approach. *The Financial Review*, 79-88.
- Gujarati, Damodar. N. (1995). *Basic econometrics*. McGraw-Hill, international Edition, Economic Series.
- Hawawini, G., C. Viallet, A. Vora. (1986, Summer). Industry influence on corporate working capital decisions, *Sloan Management Review*, 27, 4, 5-25.
- Johnson, Clive and Arun Aggarwal. (1988, December). Cash management comes to the ball. *Accountancy*, 102(1144), 81-86.
- Kamath, Ravindra. (Jan/Feb, 1989). How useful are common liquidity measures? *Journal of Cash Management*, 9(1), pp 24-28.
- Kolay, M.K. (1991). Managing working capital crisis: A system dynamics approach. *Management Decision*, 29 (5), 46-52.
- Loeser, David. (1988, November). Improving accounts receivable management. *Journal of Accounting*, 166(5), 116-117
- Miller, M. H. and D.Orr. (1966, August). A Model for the demand of money by firms. *Quarterly Journal of Economics*, 80, 413-435.
- Nicolas, Cole. (1991, Jan). When the numbers do not add up. *Director*, 44 (6), 61-68.
- Raspanti, Thomas M. (2000, Nov). What is new with trade credit insurance? An update of the U.S. market place. *Business Credit*, 102 (10), 84.

- Richards, Verlyn D. and Eugene J. Laughlin, (1980, Spring). A cash conversion cycle approach to liquidity analysis. *Financial Management*, 32-38.
- Sanger, J. S. (2001, Oct). Working capital: A modern approach, *Financial Executive*, 69.
- Schrieder, Alan J. (1988, Nov/Dec). Beyond managing cash, to managing cash flow. *Financial Executive*, 4(6), 54-57.
- Shin, Hyun-Han and Luc Soenem (1998, Fall/Winter). Efficiency of working capital management and corporate profitability. *Journal of Financial Practice and Education*, 8(2), 37-45.
- Summers, Barbara and Nicholas Wilson. (2000). Trade credit management and the decision to use factoring: An empirical study. *Journal of Business Finance and Accounting*, 27 (1-2), 37-68.
- Unsworth, Edwin (2000, Nov 27). Swiss Re, Moody's foresee growth for trade credit market. *Business Insurance*, 34(48), 15-16.

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